The manual can be summarized into **4 major technologies**:

1. HTML (Foundation)
2. CSS (Styling)
3. Bootstrap (Framework)
4. Tailwind CSS (Utility Framework)

HTML : Used to create the structure and content of web pages.(Defines what appears on a webpage.)

CSS: Used to style and format HTML elements.(Controls the appearance of a webpage.)

Bootstrap: Bootstrap is a pre-built CSS framework that helps create responsive and attractive websites quickly.(Provides ready-made components and layouts.)

Tailwind: Tailwind CSS is a utility-first CSS framework that allows styling directly through classes in HTML.(Create custom designs without writing much CSS.)

Website = HTML + CSS + Bootstrap/Tailwind

- HTML → What is displayed
- CSS → How it looks
- Bootstrap → Fast pre-designed UI
- Tailwind → Fast custom UI without writing separate CSS files

Lab 1

1. Internet

Definition

The Internet is a worldwide network of interconnected computers that communicate with each other.

Example

When you open Google or YouTube, you are using the Internet.

---

2. World Wide Web (WWW)

Definition

The Web is a collection of websites and webpages that run on the Internet.

Difference

Internet	Web
Network of computers	Collection of websites
Infrastructure	Service running on Internet

Example

Internet = Road Network

Web = Cars traveling on roads

---

3. Web Browser

Definition

Software used to access websites.

Examples

- Chrome
- Edge
- Firefox
- Opera
- Safari

Function

Converts HTML, CSS, and JavaScript into visible webpages.

---

4. Search Engine

Definition

A website that helps users find information on the Internet.

Examples

- Google
- Bing
- Yahoo
- DuckDuckGo

Difference

Browser	Search Engine
Application	Website
Chrome	Google
Firefox	Bing

Internet Protocols

5. Protocol

Definition

A set of rules that computers follow while communicating.

Example

Like grammar rules in a language.

---

6. HTTP

HyperText Transfer Protocol

Definition

Protocol used for transferring webpages between browser and server.

Example

http://example.com

Features

- Fast

- Not secure
- Data not encrypted

---

7. HTTPS

HyperText Transfer Protocol Secure

Definition

Secure version of HTTP.

Example

https://google.com

Features

- Encrypted
- Safe for passwords and payments
- Uses SSL/TLS certificate

Difference

HTTP	HTTPS
Not secure	Secure
No encryption	Encryption
Port 80	Port 443

---

8. FTP

File Transfer Protocol

Definition

Used to transfer files between computers over a network.

Uses

- Upload website files
- Download server files

Examples

- FileZilla
- WinSCP

HTTP Request and Response

9. Client

Definition

The device requesting data.

Example: Your browser

---

10. Server

Definition

Computer that provides requested data.

Example: Google's server

---

11. HTTP Request

Definition

Message sent by client to server requesting data.

Example

GET /index.html HTTP/1.1

Browser asks:
"Give me index.html"

---

12. HTTP Response

Definition

Message returned by server.

Example

HTTP/1.1 200 OK

Server says:
"Here is your page."

GET Method

Definition

Used to retrieve data from server.

Example

https://google.com/search?q=html

Characteristics

- Data visible in URL
- Faster
- Less secure

POST Method

Definition

Used to send data to server.

Example

Login Form

Username

Password

Characteristics

- Data hidden
- More secure
- Used for forms

Difference

GET	POST
Fetch data	Send data
Data in URL	Data in body
Less secure	More secure

HTTP Status Codes

200 OK

Request successful.

Example:
Website opens correctly.

---

201 Created

Resource created successfully.

Example:
Account created.

---

301 Moved Permanently

Page moved to a new URL.

---

400 Bad Request

Invalid request sent.

---

401 Unauthorized

Login required.

---

403 Forbidden

Access denied.

---

404 Not Found

Page not found.

Example:
Wrong URL entered.

---

500 Internal Server Error

Problem on server side.

---

503 Service Unavailable

Server temporarily unavailable.

Example:
IRCTC during heavy traffic.

Viva Questions from Lab 1

1. What is the Internet?
2. Difference between Internet and WWW?
3. What is a protocol?

4. What is HTTP?
5. What is HTTPS?
6. What is FTP?
7. Difference between HTTP and HTTPS?
8. What is a browser?
9. What is a search engine?
10. Difference between GET and POST?
11. What is a client?
12. What is a server?
13. What is an HTTP request?
14. What is an HTTP response?
15. What is status code 404?
16. What is status code 200?
17. What is status code 500?